

1. Problem 1

Use the Laplace Transform (LT) and its Inverse Laplace Transform (ILT) to solve the following differential equation.

i) $\ddot{y} + 2\dot{y} + y = 2, y(0) = 0, \dot{y}(0) = 0$

ii) $\ddot{y} + 2\dot{y} + y = \sin t, y(0) = 0, \dot{y}(0) = 0$

iii) $\ddot{y} - 6\dot{y} + 9y = 0, y(0) = 1, \dot{y}(0) = 1$

iv) $y^{(3)} - 3\ddot{y} - 4\dot{y} = 0, y(0) = 1, \dot{y}(0) = 1, \ddot{y}(0) = 1$

v) $\ddot{y} + 4\dot{y} + 5y = e^{-2t} \sin t, y(0) = 0, \dot{y}(0) = 0$

vi) $\ddot{y} - 4\dot{y} + 4y = 1 + \sin t + te^{2t}, y(0) = 0, \dot{y}(0) = 0$

vii) $t\ddot{y} + 2\dot{y} = 12 \ln t, y(1) = 0, \dot{y}(1) = 0$

i) ① LT (Laplace Transform)

$$\ddot{y} + 2\dot{y} + y = 2, y(0) = 0, \dot{y}(0) = 0$$

$$\mathcal{L}\{\ddot{y}\} + 2\mathcal{L}\{\dot{y}\} + \mathcal{L}\{y\} = \mathcal{L}\{2\}$$

$$s^2 Y(s) - sy(0) - \dot{y}(0) + 2[sY(s) - y(0)] + Y(s) = \frac{2}{s}$$

$$(s^2 + 2s + 1)Y(s) = \frac{2}{s}$$

$$Y(s) = \frac{2}{s(s^2 + 2s + 1)} = \frac{2}{s(s+1)(s+1)}$$

② PFD (Partial Fraction Decomposition)

$$\frac{2}{s(s+1)^2} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2}$$

$$A(s+1)^2 + B(s)(s+1) + Cs = 2$$

$$A(s^2 + 2s + 1) + Bs^2 + Bs + Cs = 2$$

$$(A+B)s^2 + (2A+B+C)s + A = 2$$

$$A = 2$$

$$A+B=0 \quad B=-2$$

$$2A+B+C=0$$

$$2(2) - 2 + C = 0$$

$$C = -2$$

$$Y(s) = \frac{2}{s(s+1)^2} = \frac{2}{s} + \frac{-2}{s+1} + \frac{-2}{(s+1)^2}$$

③ ILT (Inverse Laplace Transform)

$$y(t) = \mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{s}\right\} + \mathcal{L}^{-1}\left\{-\frac{2}{s+1}\right\} + \mathcal{L}^{-1}\left\{\frac{-2}{(s+1)^2}\right\}$$

$$y(t) = 2 - 2e^{-t} - 2t \cdot e^{-t} = 2[1 - (1+t)e^{-t}]$$

ii) ① LT:

$$\ddot{y} + 2\dot{y} + y = \sin t, \quad y(0) = 0, \quad \dot{y}(0) = 0$$

$$s^2 Y(s) + 2sY(s) + Y(s) = \frac{1}{s^2+1}$$

$$(s^2 + 2s + 1)Y(s) = \frac{1}{s^2+1}$$

$$\textcircled{2} Y(s) = \frac{1}{(s^2+1)(s^2+2s+1)} = \frac{1}{(s^2+1)(s+1)^2} = \frac{As+B}{s^2+1} + \frac{C}{s+1} + \frac{D}{(s+1)^2}$$

$$As + B(s+1)^2 + C(s+1)(s^2+1) + D(s^2+1) = 1$$

$$\text{let } s = -1 \quad 0 + 0 + D(1+1) = 1 \Rightarrow D = \frac{1}{2}$$

$$(As+B)(s^2+2s+1) + Cs^3 + Cs^2 + Cs + C + \frac{1}{2}s^2 + \frac{1}{2} = 1$$

$$\cancel{As^3} + 2\cancel{As^2} + \cancel{As} + \cancel{Bs^2} + 2Bs + \cancel{B} + \cancel{Cs^3} + \cancel{Cs^2} + \cancel{Cs} + \cancel{C} + \frac{1}{2}s^2 + \frac{1}{2} = 1$$

$$(A+C)s^3 + (2A+B+C+\frac{1}{2})s^2 + (A+2B+C)s + B+C+\frac{1}{2} = 1$$

$$B+C = \frac{1}{2}$$

$$A+C = 0$$

$$2A+B+C+\frac{1}{2} = 0$$

$$A+2B+C = 0$$

$$A = -C$$

$$2(-C) + 0 + C + \frac{1}{2} = 0 \quad -C + C + 2B = 0 \Rightarrow B = 0$$

$$A = -\frac{1}{2}$$

$$C = \frac{1}{2}$$

$$Y(s) = \frac{-\frac{1}{2}s}{s^2+1} + \frac{\frac{1}{2}}{s+1} + \frac{\frac{1}{2}}{(s+1)^2}$$

③ ILT:

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = -\frac{1}{2} \mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\} + \frac{1}{2} \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} + \frac{1}{2} \mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\}$$

$$= -\frac{1}{2} \cos t + \frac{1}{2} e^{-t} + \frac{1}{2} t e^{-t}$$

$$= \frac{1}{2} [-\cos t + (1+t)e^{-t}]$$

iii) ① LT:

$$y'' - 6y' + 9y = 0, \quad y(0) = 1, \quad y'(0) = 1$$

$$s^2 Y(s) - s y(0) - y'(0) - (6s Y(s) - y(0)) + 9Y(s) = 0$$

$$s^2 Y(s) - s - 1 - 6s Y(s) + 6 + 9Y(s) = 0$$

$$(s^2 - 6s + 9)Y(s) - s + 5 = 0$$

$$Y(s) = \frac{s-5}{s^2-6s+9}$$

② PFD:

$$\frac{s-5}{s^2-6s+9} = \frac{A}{(s-3)} + \frac{B}{(s-3)^2}$$

$$s-5 = A(s-3) + B$$

$$A=1 \quad -5 = -3A+B$$

$$s-5 = As - 3A + B \Rightarrow$$

$$B = -2$$

$$Y(s) = \frac{1}{s-3} + \frac{-2}{(s-3)^2}$$

③ ILT:

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = e^{3t} - 2te^{3t}$$

iv) ① LT:

$$y^{(3)} - 3y'' - 4y' = 0, \quad y(0) = 1, \quad y'(0) = 1, \quad y''(0) = 1$$

$$s^3 Y(s) - s^2 y(0) - s y'(0) - y''(0) - 3[s^2 Y(s) - s y(0) - y'(0)] - 4[s Y(s) - y(0)] = 0$$

$$s^3 Y(s) - s^2 - s - 1 - 3[s^2 Y(s) - s - 1] - 4[s Y(s) - 1] = 0$$

$$\cancel{s^3 Y(s)} - s^2 - s - 1 - 3\cancel{s^2 Y(s)} + 3s + 3 - 4\cancel{s Y(s)} + 4 = 0$$

$$(s^3 - 3s^2 - 4s) Y(s) - s^2 + 2s + 6 = 0$$

$$(s^3 - 3s^2 - 4) Y(s) = s^2 - 2s - 6$$

$$Y(s) = \frac{s^2 - 2s - 6}{(s^3 - 3s^2 - 4)}$$

② PFD:

$$Y(s) = \frac{s^2 - 2s - 6}{s(s+1)(s-4)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s-4}$$

$$A(s+1)(s-4) + B(s)(s-4) + C(s)(s+1) = s^2 - 2s - 6$$

$$A(s^2 - 3s - 4) + Bs^2 - 4Bs + Cs^2 + Cs = s^2 - 2s - 6$$

$$(A+B+C)s^2 - (3A+4B-C)s - 4A = s^2 - 2s - 6$$

$$A+B+C=1 \quad -3A-4B+C=-2 \quad -4A=-6 \quad A=\frac{3}{2}$$

$$B+C=-\frac{1}{2} \quad -3\left(\frac{3}{2}\right)-4B+C=-2$$

$$C=-B-\frac{1}{2} \quad -\frac{9}{2}-4B+C=-2$$

$$C=-\frac{1}{10} \quad -\frac{9}{2}-4B-B-\frac{1}{2}=-2$$

$$-5B=3$$

$$B=-\frac{3}{5}$$

$$Y(s) = \frac{\frac{3}{2}}{s} + \frac{-\frac{3}{5}}{s+1} + \frac{\frac{1}{10}}{s-4}$$

③ ILT:

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = \frac{3}{2} - \frac{3}{5}e^{-t} + \frac{1}{10}e^{4t}$$

v) ① LT:

$$\ddot{y} + 4\dot{y} + 5y = e^{-2t} \sin t, \quad y(0)=0, \quad \dot{y}(0)=0$$

$$s^2 Y(s) - sy(0) - y'(0) + 4[sY(s) - y(0)] + 5Y(s) = \frac{1}{(s+2)^2 + 1}$$

$$(s^2 + 4s + 5) Y(s) = \frac{1}{s^2 + 4s + 5}$$

$$Y(s) = \frac{1}{(s^2 + 4s + 5)^2}$$

② PFD:

$$Y(s) = \frac{A}{s - (-2+i)} + \frac{B}{[s - (-2+i)]^2} + \frac{C}{s - (-2-i)} + \frac{D}{[s - (-2-i)]^2}$$

$$Y(s) = \frac{A}{s+2-i} + \frac{B}{(s+2-i)^2} + \frac{C}{s+2+i} + \frac{D}{(s+2+i)^2}$$

$$A(s+2-i)(s+2+i)^2 + B(s+2+i)^2 + C(s+2+i)(s+2-i)^2 + D(s+2-i)^2 = 1$$

$$(s^2+4s+5)(s^2+4s+5) = s^4+4s^3+5s^2+4s^3+16s^2+20s+5s^2+20s+25 \\ = s^4+8s^3+26s^2+40s+25$$

(By using matlab)

$$Y(s) = \frac{-\frac{1}{4}i}{s+2-i} + \frac{-\frac{1}{4}}{s+2-i} + \frac{\frac{1}{4}i}{s+2+i} + \frac{-\frac{1}{4}}{s+2+i}$$

$$\textcircled{3} \text{ ILT: } -0.25ie^{(-2+i)t} + -0.25e^{(-2+i)t} + 0.25ie^{(-2-i)t} + -0.25e^{(-2-i)t}$$

$$= \frac{1}{4}e^{-2t} [-ie^{it} - e^{it} + ie^{-it} - e^{-it}]$$

$$= \frac{1}{4}e^{-2t} [-i(e^{it} - e^{-it}) - (e^{it} + e^{-it})]$$

$$= \frac{1}{4}e^{-2t} \left[2 \frac{e^{it} - e^{-it}}{2i} - 2 \left(\frac{e^{it} + e^{-it}}{2} \right) \right]$$

$$= \frac{1}{4}e^{-2t} (2\sin t - 2\cos t)$$

$$= \frac{1}{2}e^{-2t} (\sin t - \cos t)$$

Euler's identity (sine)

Euler's identity (cosine)

vi) ① LT:

$$\ddot{y} - 4\dot{y} + 4y = 1 + \sinh t + te^{2t}, \quad y(0) = 0, \quad \dot{y}(0) = 0$$

$$s^2 Y(s) - sy(0) - \dot{y}(0) - 4[sY(s) - y(0)] + 4Y(s) = \frac{1}{s} + \frac{1}{s^2+1} + \frac{1}{(s-2)^2}$$

$$(s^2 - 4s + 4)Y(s) = \frac{1}{s} + \frac{1}{s^2+1} + \frac{1}{(s-2)^2}$$

$$(s-2)^2 Y(s) = \frac{1}{s} + \frac{1}{s^2+1} + \frac{1}{(s-2)^2}$$

$$(s-2)^2 Y(s) = \frac{(s^2+1)(s-2)^2 + s(s-2)^2 + s(s^2+1)}{s(s^2+1)(s-2)^2}$$

$$Y(s) = \frac{(s^2+1)(s^2-4s+4) + s(s^2-4s+4) + s^3+s}{(s^3+s)(s^2-4s+4)(s^2-4s+4)}$$

$$Y(s) = \frac{\cancel{s^4} - \cancel{4s^3} + \cancel{s^2} - \cancel{4s} + 4 + \cancel{s^3} - \cancel{4s^2} + \cancel{4s} + s^3 + s}{s^3(s^4 - 8s^3 + 24s^2 - 32s + 16) + s(s^4 - 8s^3 + 24s^2 - 32s + 16)}$$

$$= \frac{s^4 - 2s^3 + s^2 + s + 4}{s^7 - 8s^6 + 24s^5 - 40s^4 + 40s^3 - 32s^2 + 16s}$$

Vii) ① LT:

$$t\ddot{y} + 2\dot{y} = 12 \ln t, \quad y(1) = 0, \quad \dot{y}(1) = 0$$

Euler-Mascheroni
constant

$$t[s^2 Y(s) - sy(1) - y'(1)] + 2[sY(s) - y(1)] = 12\left(-\frac{\gamma}{s} - \frac{\ln s}{s}\right)$$

2. Problem 2

Use the Laplace Transform (LT) and its Inverse Laplace Transform (ILT) to solve the following differential equation. You may retain the Laplace operator $\mathcal{L}$ and the inverse Laplace operator $\mathcal{L}^{-1}$ in the equation.

2.1 a

$$\tau \dot{y} + y = Ku(t)$$

Where τ and K are constants.

2.2 b

$$\frac{1}{\omega_n^2} \ddot{y} + \frac{2\zeta}{\omega_n} \dot{y} + y = Ku(t)$$

Where ω_n , ζ and K are constants.

2.3 c

Please decompose the obtained analytical solution into terms related only to the initial values and terms related to the external input $u(t)$.

2.4 d

We refer to the case where all initial states are zero and the input is a unit step signal as the zero-state response. Please provide the zero-state responses of the first-order system (a) and the second-order system (b).

2.1 a) $\tau \dot{y} + y = Ku(t)$

$$\tau [sY(s) - y(0)] + Y(s) = K \mathcal{L}\{u(t)\}$$

$$(\tau s + 1)Y(s) = K U(s) + \tau y(0)$$

$$Y(s) = \underbrace{\frac{K}{\tau s + 1} \cdot U(s)}_{\text{forced response}} + \underbrace{\frac{\tau}{\tau s + 1} y(0)}_{\text{free response}} = \frac{K}{s + \frac{1}{\tau}} \cdot U(s) + \frac{\tau}{s + \frac{1}{\tau}} y(0)$$

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = K \cdot \mathcal{L}^{-1}\{U(s)\} e^{-\frac{1}{\tau}t} + \tau y(0) e^{-\frac{1}{\tau}t}$$

2.2 b) $\frac{1}{\omega_n^2} \ddot{y} + \frac{2\zeta}{\omega_n} \dot{y} + y = Ku(t)$

$$\ddot{y} + 2\zeta\omega_n \dot{y} + \omega_n^2 y = K \mathcal{L}\{u(t)\} \cdot \omega_n^2$$

$$s^2 Y(s) - s y(0) - y'(0) + 2\zeta\omega_n [sY(s) - y(0)] + Y(s) \omega_n^2 = K \omega_n^2 \mathcal{L}\{u(t)\}$$

2.3 c) $(s^2 + 2\zeta\omega_n s + \omega_n^2)Y(s) = K \omega_n^2 \mathcal{L}\{u(t)\} + s y(0) + y'(0) + 2\zeta\omega_n y(0)$

$$Y(s) = \frac{K \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \mathcal{L}\{u(t)\} + \frac{(2\zeta\omega_n + s) y(0) + y'(0)}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$y(t) = \mathcal{L}^{-1}\{Y(s)\}$$

zero-state response

$$2-4 d) \quad i) \quad Y(s) = \frac{K}{\tau s + 1} \cdot \mathcal{L}\{u(t)\} = \frac{K}{\tau s + 1} \cdot \frac{1}{s} = \frac{K}{(\tau s + 1)s}$$
$$y(t) = K e^{-\frac{1}{\tau}t}$$

$$ii) \quad Y(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s}$$
$$= \frac{K\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

zero input response

$$i) \quad Y(s) = \frac{\tau}{s + 1} y(0)$$

$$y(t) = \tau y(0) \cdot e^{-\frac{1}{\tau}t}$$

$$ii) \quad Y(s) = \frac{(2\zeta\omega_n + s)y(0) + y'(0)}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} =$$

3. Problem 3

In this problem, we will briefly explore the similarities and differences between the state-space form and the input-output representation using Laplace transforms in the solution process.

3.1 a(Input/Output Representation)

Consider a system represented as:

$$\left[\left(\frac{d}{dt} + 1 \right)^2 \left(\frac{d}{dt} + 2 \right) \right] y = 1, \quad y(0) = 0, \quad \dot{y}(0) = 0, \quad \ddot{y}(0) = 0$$

where $\frac{d}{dt}$ is the differential operator. This differential equation is equivalent to:

$$y^{(3)} + 4\ddot{y} + 5\dot{y} + 2y = 1$$

Please use the Laplace Transform and its inverse to solve this differential equation.

$$y^{(3)} + 4\ddot{y} + 5\dot{y} + 2y = 1$$

$$s^3 Y(s) - s^2 y(0) - s \dot{y}(0) - \ddot{y}(0) + 4[s^2 Y(s) - s y(0) - \dot{y}(0)] + 5[s Y(s) - y(0)] + 2Y(s) = \frac{1}{s}$$

$$+ 2Y(s) = \frac{1}{s}$$

$$(s^3 + 4s^2 + 5s + 2) Y(s) = \frac{1}{s}$$

$$Y(s) = \frac{1}{s(s^3 + 4s^2 + 5s + 2)}$$

PFD :

$$Y(s) = \frac{1}{s(s+1)^2(s+2)} \quad (s^2 + 2s + 1)$$

$$Y(s) = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2} + \frac{D}{s+2}$$

$$A = sY(s)|_{s=0} = \frac{1}{2}$$

$$D = (s+2)Y(s)|_{s=-2} = -\frac{1}{2}$$

$$B = \frac{d}{ds} [(s+1)^2 Y(s)]|_{s=-1} = \frac{d}{ds} \left(\frac{1}{s^2 + 2s} \right) |_{s=-1} = 0$$

$$C = (s+1)^2 Y(s)|_{s=-1} = -1$$

$$Y(s) = \frac{\frac{1}{2}}{s} + \frac{-1}{(s+1)^2} + \frac{-\frac{1}{2}}{s+2}$$

ILT :

$$y(t) = \frac{1}{2} - t \cdot e^{-t} - \frac{1}{2} e^{-2t}$$

3.2 b(State-Space Representation)

Consider the same system as in part a, whose state-space representation is:

$$\begin{cases} x_1 = y \\ x_2 = \dot{y} \\ x_3 = \ddot{y} \end{cases}$$

Let $X = [x_1, x_2, x_3]^T$, $Y = [y]$, and $U = [u]$.

Thus, it can be written in the form of a system of linear differential equations:

$$\begin{cases} \dot{X} = AX + BU \\ Y = CX + DU \end{cases}$$

The key to solving this system of linear differential equations lies in solving the equation $\dot{X} = AX + BU$. This is a first-order linear ordinary differential equation system, and common methods for solving it include:

1. Decompose the matrix A into $A = P^{-1}JP$, where J is the Jordan canonical form and P is the matrix of generalized eigenvectors.
2. Make the transformation $Z = PX$, then the original equation becomes $\dot{Z} = JZ + (PB)U$.
3. Solve by independently calculating variables or using the matrix exponential method.

In this problem, you are not required to solve this system of linear differential equations. The tasks you need to complete in this question are:

1. Based on the system differential equation in 3.a, find the coefficient matrices A , B , C , and D in the state-space representation.
2. Decompose the matrix A into its Jordan canonical form; you do not need to find the matrix P . (This process is rather cumbersome and you may use MATLAB, Mathematica, or other tools to complete it.)
3. Compare the eigenvalues in the Jordan canonical form with the solutions of the differential equation in 3.a. What do you observe?

$$1- \quad x_1 = y \quad x_2 = \dot{y} \quad x_3 = \ddot{y}$$

$$\dot{x}_1 = x_2 = \dot{y}$$

$$\dot{x}_2 = x_3 = \ddot{y}$$

$$\dot{x}_3 = \ddot{y} = 1 - 4\ddot{y} - 5\dot{y} - 2y = 1 - 4x_3 - 5x_2 - 2x_1$$

$$\dot{X} = AX + BU$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -5 & -4 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = x_1$$

$$[y] = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} \cdot u$$

Jordan canonical of A (Matlab)

$$J^2 = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

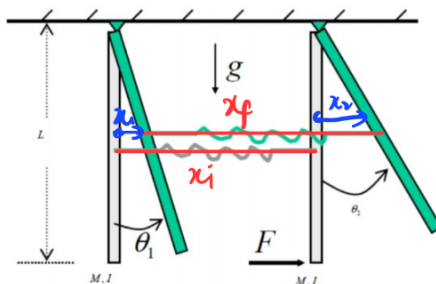
Matrix of generalized eigenvectors:

$$\begin{bmatrix} -2 & -1 & -1 \end{bmatrix}$$

3.3 c Simple Application

This problem does not restrict the solving method, but you might consider how to express the answer more concisely.

Consider the double pendulum shown in the figure:



where the two pendulums have equal mass M and moment of inertia I (with respect to the joint) and are connected at the center through a spring with a stiffness of K and a damping coefficient of B_s . The spring is un-stretched when the two pendulums are at the downward positions. At time zero, a horizontal force F is applied to the tip of the second pendulum as shown. Model the system through the following procedure:

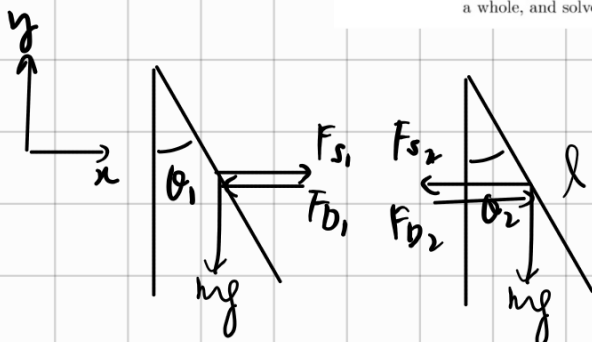
Assume all angles are small, which means you can make the following approximations: $\sin(\theta) \approx \theta$, $\cos(\theta) \approx 1$.

1. All initial states are zero.

Attempt to establish differential equations and solve for the variation of θ_1 over time. The order of the differential equation is relatively high; you may retain the Laplace operator $\mathcal{L}$ and the inverse Laplace operator $\mathcal{L}^{-1}$ in the final expression.

Tips: There are three ways of thinking for your reference, but you can also propose your own method.

1. List all differential equations and simplify them into the form where F is the input and θ_1 is the output. Then use the Laplace transform to solve.
2. (Transfer Function) List all differential equations, use the Laplace transform to convert the differential equations into algebraic equations (transfer function form), solve them simultaneously, and then perform the inverse transform.
3. Treat the entire system as a two-degree-of-freedom system, write matrix equations by considering it as a whole, and solve the matrix equations using the Laplace transform.



$$\begin{aligned} x &= x_f - x_i & x_f \text{ (final length)} \\ x_f &= x_i - x_1 + x_2 & x_i \text{ (initial length)} \\ x_1 &= \frac{l}{2} \cos \theta_1 \approx \frac{l}{2} \theta_1 \\ x_2 &= \frac{l}{2} \theta_2 \end{aligned}$$

$$x = x_f - x_i = x_2 - x_1 = \frac{l}{2} (\theta_2 - \theta_1)$$

$$\begin{aligned} J \ddot{\theta}_1 &= (F_{s1} - F_{b1}) \cdot \frac{l}{2} \cos \theta_1 - \frac{1}{2} m g l \sin \theta_1 \approx \frac{1}{2} [(F_{s1} - F_{b1}) l \theta_1 - m g l \theta_1] \\ J \ddot{\theta}_2 &= (F_{b2} - F_{s2}) \cdot \frac{l}{2} \cos \theta_2 - \frac{1}{2} m g l \sin \theta_2 \approx \frac{1}{2} [(F_{b2} - F_{s2}) l \theta_2 - m g l \theta_2] \end{aligned}$$

$$F_{S1} = K \cdot x = \frac{k_l}{2} (\theta_2 - \theta_1) = F_{S2}$$

$$F_{D1} = B \cdot \dot{x} = B (\dot{\theta}_2 - \dot{\theta}_1) = F_{D2}$$

All initial conditions are 0.